

HW6

Due on 11/25/18

(Total: 80 points)

Self-study problems:

Example problems for Ch. 4: 4.8, 4.10, 4.12 (for midterm study).

Example problems for Ch. 5: 5.6, 5.8, 5.13 (for midterm study).

Example problems for Ch. 6: 6.6, 6.7, 6.9 (for midterm study).

HW problems:

Ch5: 5.5, 5.9(b), 5.14, 5.22(a)

Ch6: 6.3, 6.6(b), 6.7(a), 6.9(a)

5.24. For a device operating in the subthreshold region, we have

$$I_D = \mu C_d \frac{W}{L} V_T^2 \left(\exp \frac{V_{GS} - V_{TH}}{V_T} \right) \left(1 - \exp \frac{-V_{DS}}{V_T} \right) \quad (5.64)$$

- (a) If the device is in the deep triode region, $V_{DS} \ll V_T$. Using $\exp(-\epsilon) \approx 1 - \epsilon$, determine the on-resistance.
- (b) If the device is in saturation, $V_{DS} \gg V_T$. Compute the transconductance.
- (c) Find the relation between $g_{m,B}$ and $R_{on,R}$ in Fig. 5.43(d) using the above results.

5.25. Determine the corner frequency resulting from C_{in} in Fig. 5.47(d). For simplicity, assume C_1 is a short circuit.

Fig 5.43(d):